

Spectroscope

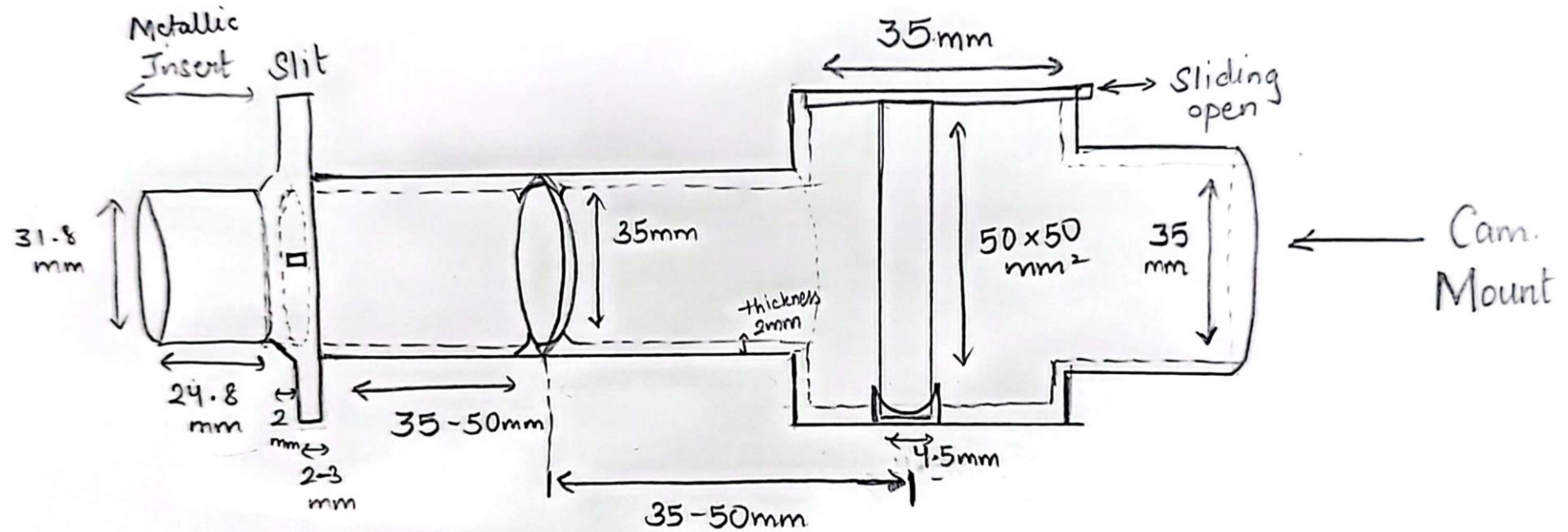
Project Brief:

- The project aims to design and build a low-cost spectroscope that can analyze light from celestial and terrestrial sources.
- It uses a slit, collimating lens, collimator, diffraction grating, and imaging sensor (QHYCCD CMOS camera) for spectrum acquisition.
- The captured spectra are processed using Python (Astropy & image analysis libraries) to extract key features like emission/absorption lines.

Outcomes:

- Capture stellar spectra (e.g., Hydrogen Balmer lines, Sodium D, Calcium H&K).
- Detect emission lines from nebulae like $H\alpha$ and [OIII].
- Measure Doppler shifts of bright celestial objects.
- Look at absorption bands in bright planets (e.g., methane in Jupiter/Saturn).
- Study light pollution and airglow lines.
- Do basic star classification by spectral type.

Model Layout Diagram:



Hardware Components:

- Adjustable slit 10-50micrometers
- Plano-convex lens- Dia=35mm, FL= less than 100mm
- Transmissive Diffraction Grating- 1000lines/mm
- 3D printed casing
- QHYCCD -QHY5III85
- Mobile phone mount

Data Flow & Tech Stack:

- CMOS Camera captures spectrum at telescope focus.
- Capture Software (SharpCap) saves images in FITS/TIFF format.
- Data Transfer via USB to PC.
- Pre-processing (bias/dark correction, calibration).
- Analysis & Visualization in Python / DS9.
- Final Results → Identify spectral lines, classify stars, detect planetary/nebular features.

Observation Challenges:

- Defect in mount locking can affect the observation, it makes the observation very difficult in long exposures.
- CMOS camera noise limits faint object detection without cooling.
- Low-cost optics: diffraction grating and lenses may introduce aberrations.
- Light pollution: strong interference for emission/absorption line studies in urban skies.

Project Budget

- Slit



3Dprinted / metal frame+knob
=Rs 200

Project Budget

- Plano –convex lens Dia – 35 mm Dia and 70 mm Focal length

<https://www.edmundoptics.in/p/38326ink/39028>

Price: **Rs. 5895**



Project Budget:

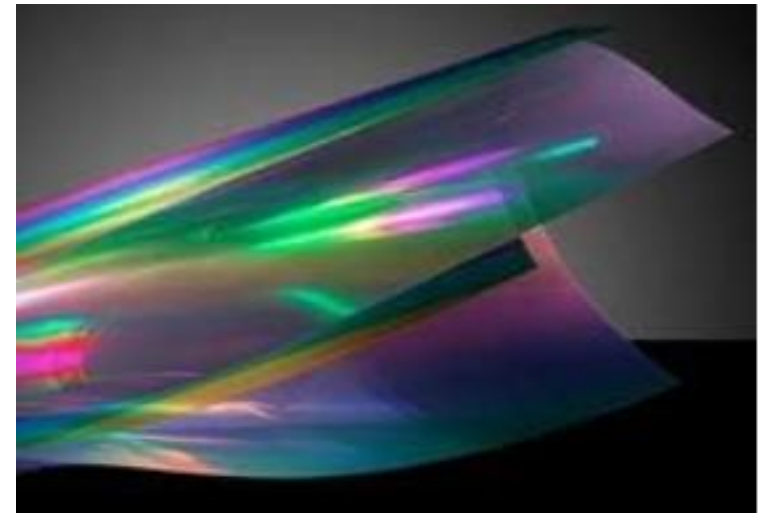
- Transmissive Diffraction Grating Sheet Single Axis Size 6" x 12" (15 x 30 cm) - 1 Piece (1000lines/mm) - <https://amzn.in/d/0Fefu8l>

Price: **Rs.1450**

Optional - <https://amzn.in/d/16rz6nO>

Size : 60X45X4mm
50X50X4mm

Price: **Rs.800**



Project Budget

- 3-D printing will cost around **Rs.1000**
- **Total cost is around: Rs. 9000 (including misc)**

Future Improvements:

- Upgrade to a higher resolution diffraction grating for finer spectral details.
- Use a cooled CCD/CMOS camera to reduce noise in faint object spectra.
- Develop a real-time spectrum extraction software pipeline
- Extend wavelength coverage to near-IR / UV regions.
- Integrate with online databases (e.g., SIMBAD, NIST lines) for automated line identification.